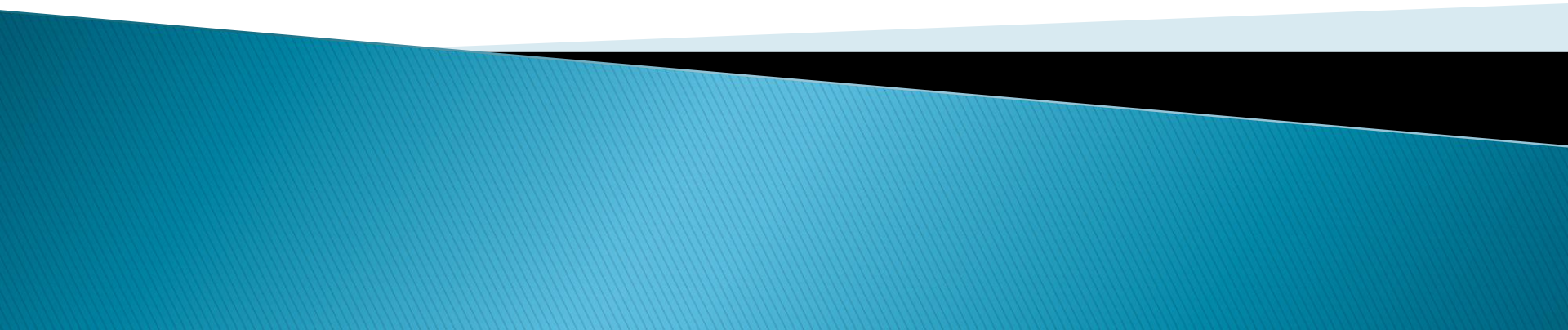


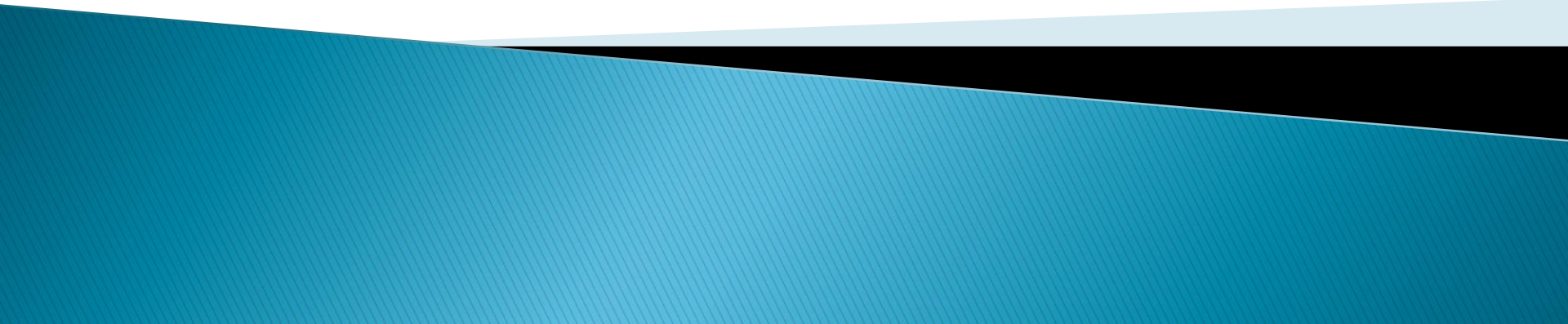
CYBER SECURITY PRINCIPLES

**BY
MERREYLYNE PRAIZE-ANYA**

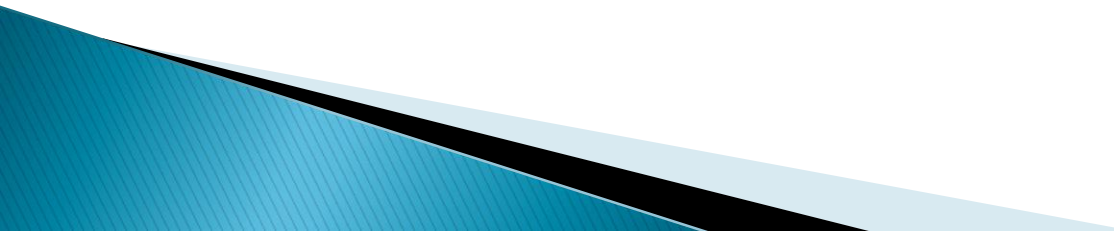


MODULE 1

PERSPECTIVES AND IMPACT OF CYBERSECURITY



OUTLINE

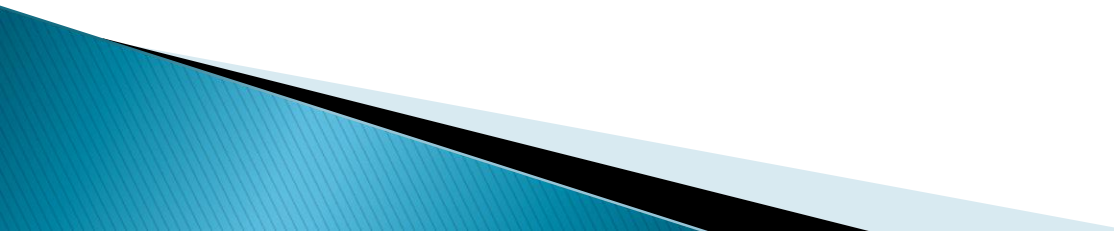
- ▶ Evolution of cybersecurity: global and organizational perspectives
 - ▶ Social, economic, and political impacts of cyber threats
 - ▶ Ethical considerations and responsible use of technology
 - ▶ Case studies: major Cybersecurity incidents and their consequences
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Introduction

- ▶ Cyber security is the most concerned matter as cyber threats and attacks are overgrowing. Attackers are now using more sophisticated techniques to target the systems. Individuals, small-scale businesses or large organization, are all being impacted. So, all these firms whether IT or non-IT firms have understood the importance of Cyber Security and focusing on adopting all possible measures to deal with cyber threats.
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Introduction

Cyber security is primarily about people, processes, and technologies working together to encompass the full range of threat reduction, vulnerability reduction, deterrence, international engagement, incident response, resiliency, and recovery policies and activities, including computer network operations, information assurance, law enforcement, etc."



What is Cyber Security



Definition:

1. Cyber security is the body of technologies, processes, and practices designed to protect networks, computers, programs and data from attack, damage or unauthorized access. □

- The term cyber security refers to techniques and practices designed to protect digital data. □
- The data that is stored, transmitted or used on an information system.

What is Cyber Security

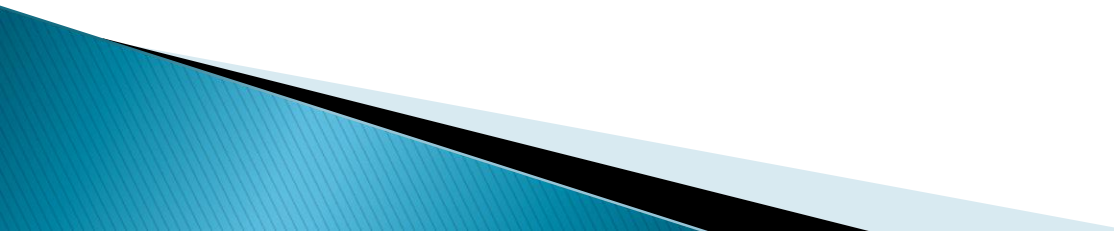


Definition

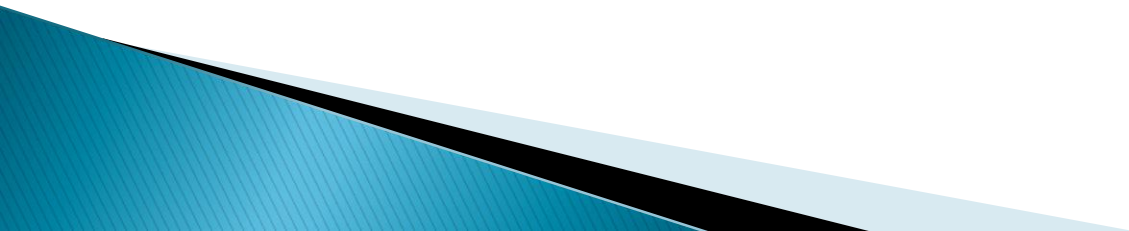
2. Cyber security is the protection of Internet-connected systems, including hardware, software, and data from cyber attacks. It is made up of two words one is cyber and other is security.

- ❑ Cyber is related to the technology which contains systems, networks and programs or data.
- ❑ Whereas security related to the protection which includes systems security, network security and application and information security

Why is cyber security important

- Cyber attacks can be extremely expensive for businesses to endure.
 - Data breach can also inflict untold reputational damage.
□
 - Cyber-attacks these days are becoming progressively destructive.
 - Cybercriminals are using more sophisticated ways to initiate cyber attacks.
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Evolution of cybersecurity: global and organizational perspectives



Social, economic, and political impacts of cyber threats

Cyber threats compromise the **Confidentiality, Integrity, and Availability (CIA)** of systems, leading to severe cascading failures across society, the economy, and the political landscape.



. Economic Impact

Area of Impact	Explanation	Key Threats
Direct Costs	Tangible, immediate expenses following an attack.	Ransomware: Payments made to decrypt systems.
	Theft of Funds: Illicit financial gains by criminals.	Data Breaches: Costs for forensic analysis, legal fees, and regulatory fines (e.g., GDPR).
Indirect Costs	Longer-term, systemic losses that are harder to quantify.	Intellectual Property (IP) Theft: Loss of trade secrets and competitive advantage through espionage.
	Stock Market Reaction: Share price drops due to loss of investor confidence and damaged reputation.	Supply Chain Disruption: Attacks on a single company (e.g., a software vendor) cripple multiple dependent firms.
	Increased Security Costs: Continuous high investment required for new defensive measures, diverting capital from innovation.	

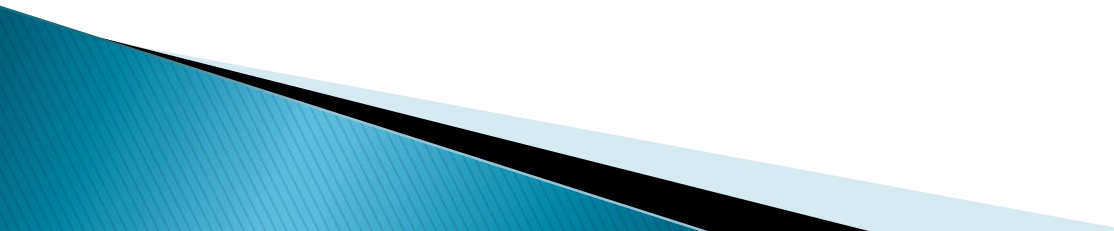
Political Impact

Area of Impact	Explanation	Key Threats
Cyber Espionage	State-sponsored actors infiltrate government and defense networks to steal sensitive information.	Stealing classified documents, military plans, and diplomatic negotiation details.
Interference in Democracy	Attacks aimed at undermining the legitimacy of democratic processes.	Election Tampering: Hacking voter registration lists or ballot systems; Disinformation Campaigns that manipulate public opinion to sway election outcomes.
Critical Infrastructure Sabotage	Attacking systems that provide essential public services.	Targeting the Power Grid, water treatment facilities, or major transport hubs to cause widespread disruption and panic (a form of cyberwarfare).
Erosion of Trust	Damage to public faith in government institutions, agencies, and their ability to protect the nation and its citizens.	High-profile breaches of government employee data or defense contractors.

3. Social Impact

Area of Impact	Explanation	Key Threats
Disruption to Daily Life	Attacks that compromise essential systems people rely on.	Disruption of healthcare systems (preventing access to patient records/care) or financial services (locking individuals out of bank accounts).
Loss of Privacy/Identity Theft	The unauthorized exposure of personal, sensitive data.	Compromise of consumer data (e.g., credit card numbers, health records) leading to fraud, emotional distress, and anxiety.
Digital/Social Divide	Individuals with lower socio-economic status or digital literacy are more vulnerable to scams and threats, worsening existing inequalities.	Phishing Attacks and scams targeting vulnerable populations.
Psychological Impact	Widespread anxiety and fear resulting from a constant state of insecurity.	The public's loss of confidence in online technology and the safety of the internet.

Ethical considerations and responsible use of technology

- ▶ Technology, while offering immense opportunities for progress and problem-solving, also presents complex ethical dilemmas. Responsible use requires proactive consideration of potential harms, equity, autonomy, and the long-term societal impact of technological advancements. This involves anticipating consequences, establishing guidelines, and fostering a culture of accountability among developers, users, and policymakers.
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Core Ethical Principles in Technology

- ▶ Several foundational ethical principles underpin discussions about responsible technology:

1. Beneficence & Non-Maleficence (Do Good & Do No Harm):

Beneficence: Technology should be designed and used to promote well-being, improve quality of life, and contribute positively to society.

Non-Maleficence: Avoid causing harm, directly or indirectly, through technology. This includes physical, psychological, social, economic, and environmental harm.

- *Examples:* Developing AI for medical diagnosis (beneficence) vs. creating autonomous weapons systems (potential maleficence).

Core Ethical Principles in Technology

2. Autonomy:

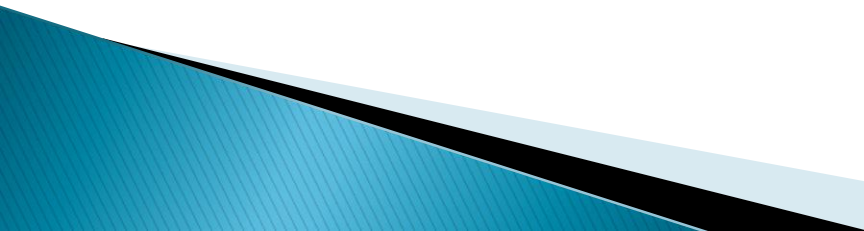
- ▶ Respect for individual choice, control, and self-determination regarding technology's impact on their lives.

- ▶ Ensuring informed consent for data collection, providing meaningful control over personal information, and avoiding manipulative or coercive design.

- ▶ *Examples:* User control over privacy settings, opting out of tracking, freedom from algorithmic bias that limits choices.

Core Ethical Principles in Technology

3. Justice & Fairness:

- ▶ Ensuring equitable access to technology's benefits and minimizing the exacerbation of existing social, economic, or political inequalities.
 - ▶ Addressing algorithmic bias that can lead to discriminatory outcomes against marginalized groups.
 - ▶ *Examples:* Bridging the digital divide, fair allocation of AI-driven resources (e.g., credit, employment), non-discriminatory facial recognition.
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Core Ethical Principles in Technology

4. Accountability & Transparency

Identifying who is responsible when technology causes harm and establishing mechanisms for redress.

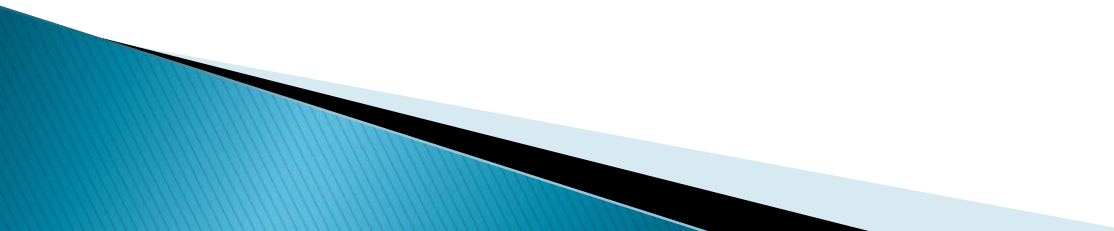
- ▶ Making technological processes (especially AI algorithms) understandable and explainable to relevant stakeholders.

Examples: Clear responsibility for autonomous vehicle accidents, explainable AI (XAI) for critical decision-making systems.

Key Areas of Ethical Concern & Responsible Use

▶ A. Data Privacy and Security

Ethical Concern: Mass collection, storage, and processing of personal data raise concerns about surveillance, re-identification, and misuse. Data breaches expose individuals to identity theft and harm.



Key Areas of Ethical Concern & Responsible Use

► **Responsible Use:**

- **Privacy by Design:** Incorporating privacy protections into the core architecture of systems.
- **Data Minimization:** Collecting only necessary data and deleting it when no longer needed.
- **Strong Security Measures:** Protecting data from unauthorized access, loss, or damage.
- **Informed Consent:** Clearly communicating data practices and obtaining explicit user permission.
- **User Control:** Providing users with tools to manage their data and privacy settings.
- *Regulations:* GDPR, CCPA are examples of legal frameworks reflecting these principles.

Key Areas of Ethical Concern & Responsible Use

▶ **B. Artificial Intelligence (AI) and Machine Learning (ML)**

Ethical Concern:

- **Bias:** Algorithms trained on biased data can perpetuate or amplify existing societal inequalities (e.g., in hiring, lending, criminal justice).
- **Accountability:** Difficulty in assigning blame when autonomous AI systems cause harm.
- **Transparency/Explainability (Black Box Problem):** Complex AI models can make decisions that are difficult for humans to understand or audit.
- **Job Displacement:** Automation replacing human labor.
- **Autonomous Weapons:** Ethical dilemmas of machines making life-or-death decisions without human intervention.

Key Areas of Ethical Concern & Responsible Use

- ▶ **Responsible Use:**
 - **Fairness Audits:** Regular testing for algorithmic bias and active steps to mitigate it.
 - **Human Oversight:** Ensuring human control and intervention, especially in critical AI applications.
- ▶ **Explainable AI (XAI):**
Developing methods to make AI decisions more interpretable.
- **Ethical AI Review Boards:**
Implementing internal and external review processes for AI development.
- **Impact Assessments:**
Proactively evaluating the social and economic impact of AI deployments.

Key Areas of Ethical Concern & Responsible Use

► C. Social Media and Digital Communication

Ethical Concern:

Misinformation/Disinformation: Spread of false information, impacting public discourse and democracy.

Echo Chambers &

Polarization: Algorithms reinforcing existing beliefs, leading to fragmentation.

Mental Health: Impact on self-esteem, anxiety, and addiction, particularly among youth.

Cyberbullying &

Harassment: Online abuse and intimidation.

Privacy Erosion: Data collection for targeted advertising and potential surveillance.

Key Areas of Ethical Concern & Responsible Use

▶ C. Social Media and Digital Communication

Responsible Use:

Platform Moderation:

Implementing policies and tools to combat hate speech, misinformation, and harassment.

Transparency in Algorithms:

Informing users about how content is recommended.

User Education: Promoting digital literacy and critical thinking skills.

Safeguards for Minors:

Protecting children and adolescents from harmful content and interactions.

Key Areas of Ethical Concern & Responsible Use

► D. Emerging Technologies (e.g., Biotechnology, Gene Editing, VR/AR, Quantum Computing)

Ethical Concern:

Gene Editing (CRISPR):

Designer babies, unintended consequences for the gene pool, access inequality.

Neurotechnology: Brain-computer interfaces raising privacy concerns about mental data, cognitive enhancement, and consent.

Virtual/Augmented Reality:

Potential for psychological impact, addictive behaviors, and blurring lines between reality and simulation.

Quantum Computing: Potential to break current encryption, necessitating new security paradigms.

Key Areas of Ethical Concern & Responsible Use

► **Responsible Use:**

Precautionary Principle:

Taking preventative action in the face of uncertainty regarding potential harm.

Public Discourse:

Fostering open discussions involving diverse stakeholders (scientists, ethicists, general public) to shape norms and regulations.

International Collaboration:

Developing global standards for technologies with far-reaching impacts.

Ethical Guidelines for Research: Establishing strict ethical review processes for cutting-edge scientific and technological research.

Stakeholders in Responsible Technology

- ▶ **Technology Developers & Companies:** Bear primary responsibility for ethical design, development, and deployment.
- ▶ **Governments & Regulators:** Create laws, policies, and oversight mechanisms (e.g., data protection laws, AI regulations).
- ▶ **Academics & Researchers:** Identify ethical problems, develop ethical frameworks, and conduct independent analysis.
- ▶ **Civil Society & NGOs:** Advocate for public interests, raise awareness, and hold tech companies accountable.
- ▶ **Educators:** Foster digital literacy and ethical reasoning in future generations.
- ▶ **Individual Users:** Have a role in being informed, using technology responsibly, and demanding ethical products.

Case studies: Two Major UK Cyber Attacks in 2025

- ▶ The year 2025 has seen an acceleration of sophisticated threats, particularly those targeting critical supply chains and high-profile consumer brands.
- ▶ **1. The Large-Scale Retail Sector Ransomware and Data Theft**
- ▶ This attack vector involved high-profile criminal groups (such as the **DragonForce** or **Scattered Spider** affiliates) targeting large UK retailers, often via a weakness in a **third-party supplier** (a supply chain attack)

Studies: Two Major UK Cyber Attacks in 2025

Attack Target	Key Attackers/Method	Consequences
Major Retailers (e.g., Marks & Spencer, Co-op)	Ransomware & Data Exfiltration: Attackers breached outsourced IT systems, gaining access to internal networks.	Financial Loss & Operational Chaos: For some, this led to online sales outages lasting weeks (costing hundreds of millions of pounds), forcing stores into manual operations , and causing supply chain breakdowns due to inaccessible systems.
Legal Aid Agency (LAA)	Data Breach/Extortion: A large breach compromising public sector data.	Erosion of Public Trust & Fraud Risk: Exposed sensitive personal data of over 2 million individuals, including criminal records and financial details, raising major concerns about identity fraud and extortion.

Consequences

- ▶ **Massive Financial Impact:** Direct costs from operational downtime, ransom demands (if paid), and forensic investigation. For some firms, the financial impact has run into the hundreds of millions.
 - ▶ **Reputational Damage:** Significant **loss of customer trust** due to the exposure of sensitive personal information.
 - ▶ **Regulatory Fines:** Companies face hefty fines from the Information Commissioner's Office (ICO) for failing to protect customer data adequately.
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